

Absa Project

Group 10

Introduction

In response to a critical production challenge faced by a prominent wine manufacturer, our consulting firm specializing in data-driven solutions is set to embark on a strategic assignment. The manufacturer, seeking to enhance its production strategy, has approached us for assistance in optimizing the quantity of cases for different wine variants, aligning production with market demand, and ensuring efficient resource utilization.

This multifaceted problem requires a nuanced approach, blending predictive modeling and data analysis. Our task involves predicting the quantity of cases ordered and assessing the quality of wine variants, encompassing both regression and classification aspects. Linear regression will be employed to forecast sales quantity, while logistic regression will categorize wines as good or bad based on their chemical properties.

The assignment is underpinned by two datasets, each representing a specific wine variant – red and white. These datasets encompass 13 columns, detailing the chemical properties of the wines, customer-rated quality, and the quantity sold. This structured dataset provides a rich foundation for exploring the intricate relationships between chemical attributes, perceived quality, and commercial success.

Our objectives are: **Data Handling and Quality Assurance:** Apply rigorous data handling techniques to ensure the integrity and completeness of the provided wine datasets. This involves addressing missing values, outliers, and ensuring data reliability.

Regression Analysis: Develop and apply regression analysis techniques to predict the quantity of cases ordered and the quality of wine based on relevant features. This analytical approach will provide valuable insights into the factors influencing sales and customer perceptions.

Exploratory Data Analysis (EDA): Conduct a thorough EDA to unravel patterns, distributions, and correlations within the dataset. By delving into the interplay between chemical attributes, perceived quality, and commercial success, we aim to unearth actionable insights for strategic decision-making.

Model Evaluation and Interpretation: Assess the performance of the regression models and interpret findings, shedding light on how the quality of wine influences sales. This step is crucial for distilling actionable recommendations that can guide the manufacturer's production strategy.

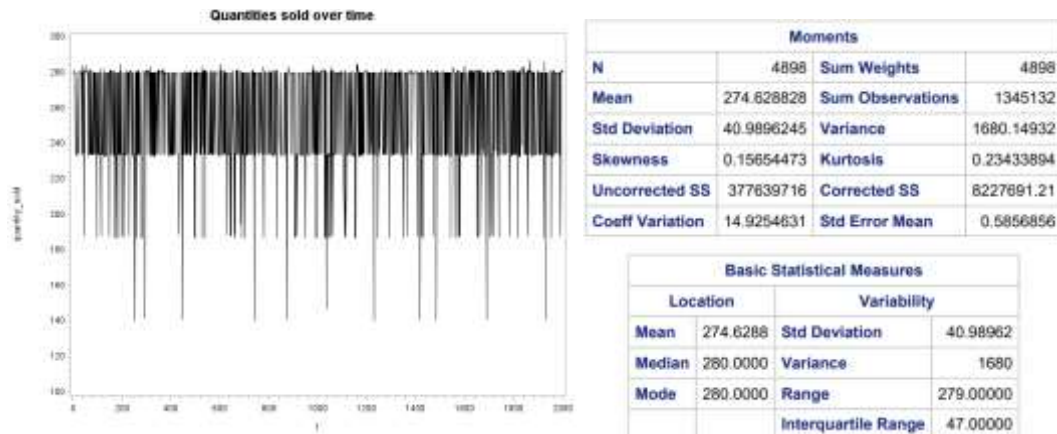
By achieving these goals, we hope to provide a comprehensive solution that not only solves the current production issue but also gives the wine producer the data-driven insights they need to navigate the market's intricacies and maintain an advantage over competitors.

Red and White wine analysis

EDA:

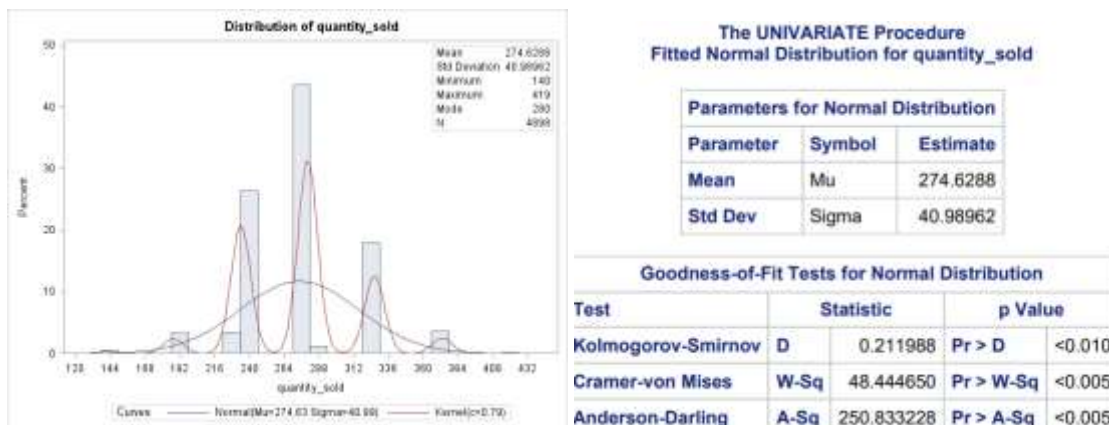
For both red wine and white wine we got the following results

The data does not have any missing values and we can't say anything about the presence of outliers because we no information about the distribution followed by the data. We used the data as it is



From the above descriptive statistics it follows that the redwine data is positively skewed and does not have a clear pattern, this follows from the above plot.

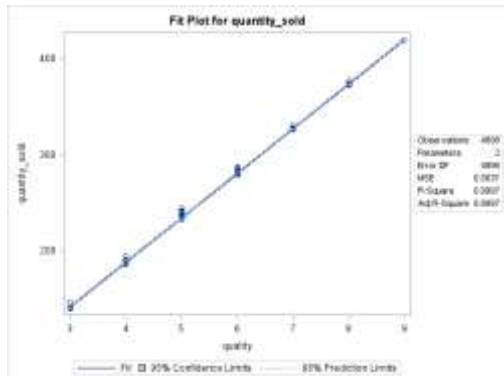
Testing whether the wine's data is normally distributed:



The above plot and goodness of fit results indicate that wine's data does not follow a normal distribution. As it also evident that all the p values of goodness of fit results are < 1% (significance level)

Now fitting a model: (results are same for both red & white wine)

Using a quality as an input parameter of our model we get the following results



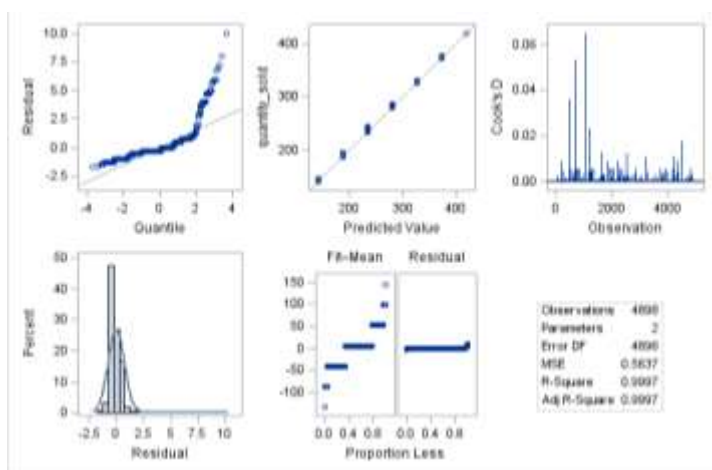
Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	2.62974	0.07201	36.52	<.0001
quality	1	46.27480	0.01211	3819.76	<.0001

Defining parameters:

Quantity sold = y and quality = x

$$Y = 2.62974 + 46.27480x$$

Evaluation of the model

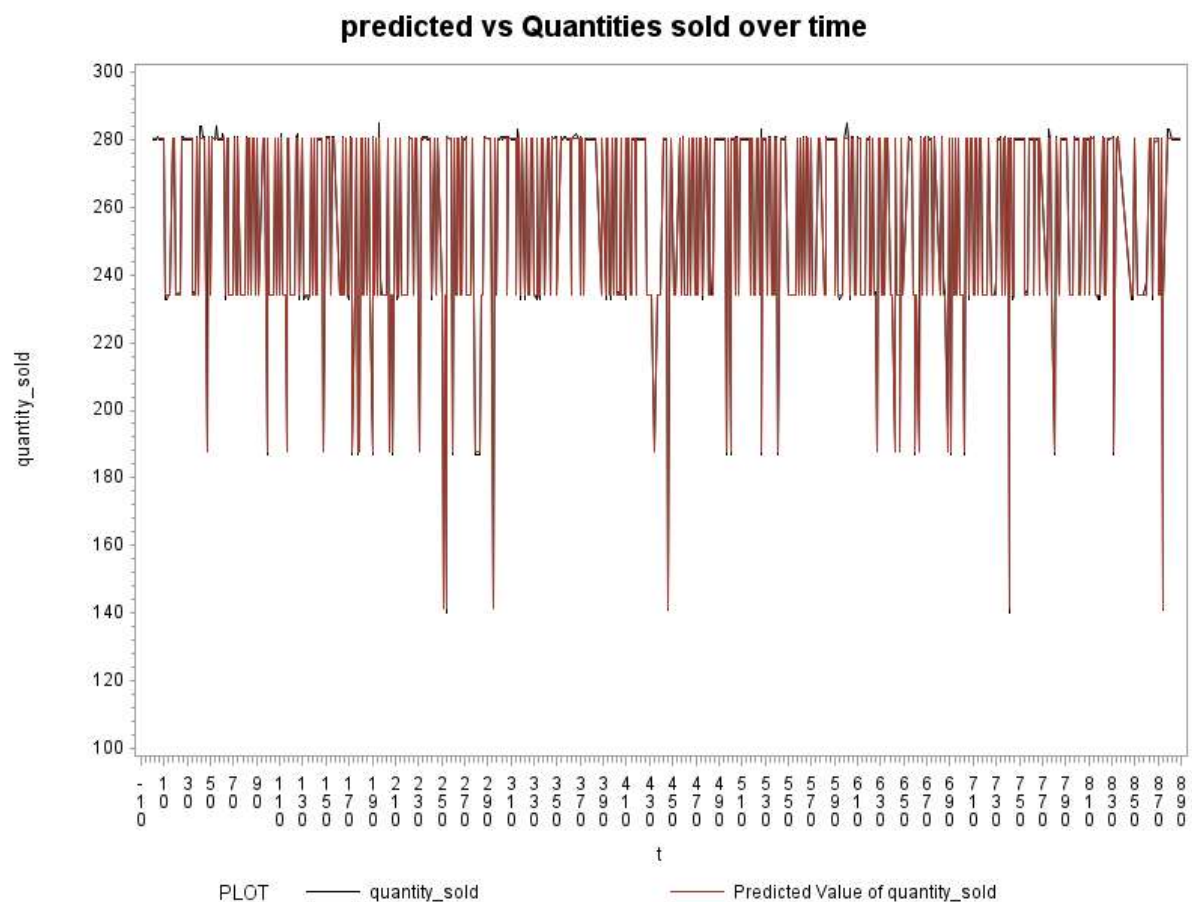


This model explain 99.97% of the wine's variation.

The model has some drawbacks such as:

- having both parameters significantly not different from zero (this follows from p values in the parameter estimate being $< 1\%$ significance level).
- The residuals are not normally distributed (this follows from the above residual histogram and qq-plot)
- A desired or perfect model must possess normally distributed residuals with mean 0, this would be an indication that fitting model has an average 0 errors. Now that residuals of our fitted model are not normally distributed, this is an indication that at some time the model will make errors.

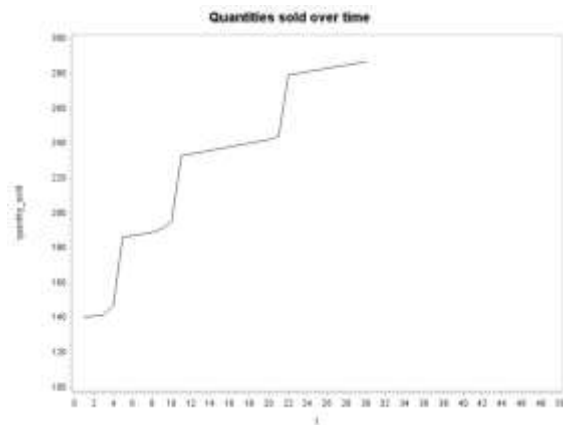
Comparing fitted model with the original data



As mentioned above the model explains almost 100% of redwine variation.

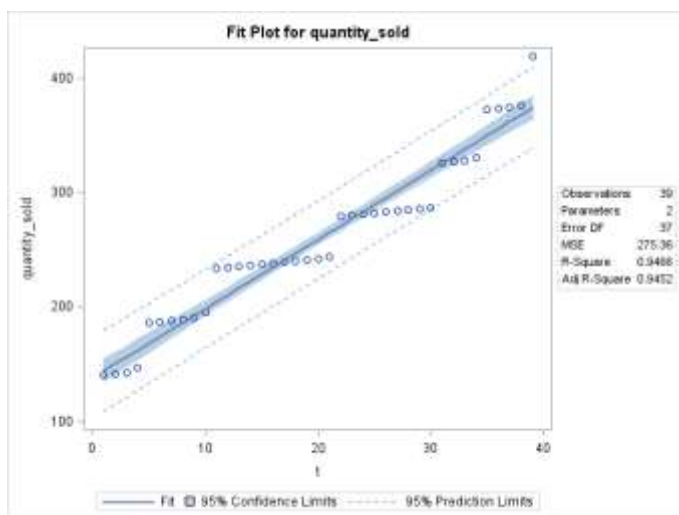
Another way would be to look at unique quantities sold overtime to get some insights:

Please note for both redwine and whitewine we got the following results



the distinct values of quantity sold over time seem to be following an increasing trend overtime

fitting a model we get these results



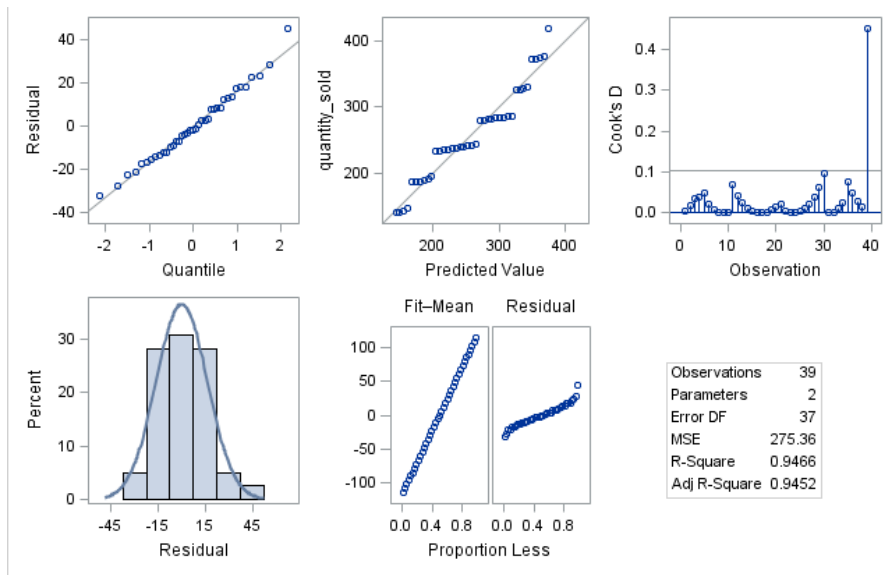
Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	137.94602	5.41822	25.46	<.0001
t	1	6.04757	0.23610	25.61	<.0001

Defining parameters

t: uniti time

y: Quantity sold (unique)

$$y = 137.94602 + 6.04757t$$



This model explains 95% of variation and that residuals are normally distributed, this is an indication that this is a good model for the unique quantities sold

Impact of wine quality on sales

Overall, a positive relationship between wine quality and sales signifies that quality is an essential driver of consumer behavior and purchasing decisions in the wine market. As a result the factory should Promote High-Quality Offerings.

Determine how many wines are good and how many are bad

This was done by creating variable called quality_category with value 1 if quality is > 5 and 0 otherwise.

We then got the following results for both models

Quantities sold over time
The FREQ Procedure

quality_category	Frequency	Percent	Cumulative Frequency	Cumulative Percent
0	1640	33.48	1640	33.48
1	3258	66.52	4898	100.00

We have about 33.48% of wines produced which are regarded as being bad wines

